This document explains how the Space Game Works

Enemies

Behavior Definition

In order to simulate bees, I made the enemies first swarm the tower they are targeting, for example first they will zoom to a tower but once in a specific range will move around it in a circle. To do this I utilized the built-in path finding algorithm and tell the pathfinder to get the next point around a center (the tower) which simulates moving in a circle. After a timer it will shoot a laser at the target tower lowering the towers health.

Targeting Types

Random: chooses the a random target in the group list of all tower nodes

Power: chooses a random target in the power group

Defense: chooses a random target in the defense group

Power

Power generation works via a graph system, the main mechanics are divided between 3 main files, global.gd and tower.gd. Each respectively governs power for towers and the global space. The principle of the power system is that each tower “requests” power when it is needed. For example if a miner wants to mine an asteroid it needs to ask for power first. This request will then be received by the power group it is in, if there is enough power in that group the miner will be allowed to perform its action, if not it will be unable to and request again.

Tower.gd (Power Generation)

Power Delay – when the tower is cooling down and doesn’t need to request more power, if modified the node will request power more often

Setting Groups

Each tower belongs to a group of other towers that it is connected to. If you have a graph of 3 towers they will all belong to a group of for example group 1. If split they will form two groups, more on this explained in global as set_group in tower is only called when it is placed

The main mechanic that is happening is we are exploiting godot’s group mechanic, This mechanic lets you put nodes in ‘groups’ for naming schemes, however it also allows you to get all towers in a group allowing us to put towers in power ‘buckets’

Global.gd (Main Scene Managment)

This is the file that manages where the power goes and what happens when a tower is destroyed.

Remove Node from group is the main mechanic that global handles and the most complex one. In order to remove the tower neatly we use a DFS search to find the tower we need to remove then systematically rewire all the connections that tower had. For example if that tower was connected to a node and that node was connected to a miner. We need to remove tower from group and also verify is no longer connected to a dangling reference of the tower we are removing. We have to do this because the power group does not contain any information regarding whats connected to what, it only shows what towers are in what group. It is also important to note that during the DFS we label any towers that will no longer be connected. These towers we will assign to a different group. This prevents a tower from requesting power from a group it no longer has any connections to.

Distributing Power

This is relatively simple, when requested a tower supplies what group it belongs to, its reference id, and also how much tower is needed. If there is enough power in the group globals will allow the tower to perform its do_process function. All towers have this function to make sure this function in globals will never attempt to call a function that doesn't exist.

Group_changed_adjust_max power

This function adjusts the maximum power a group can hold. Every tower contains a power bank, explained further in the tower section.

Towers

Every tower contains multiple functions they all have access to. These are contained in tower.gd.

Tower.gd has 4 signals all towers have access to:

draw_laser (draws a laser targeting a position, global as two towers utilize this function)

draw_hover_line (draws the lines shown when you are placing the tower)

draw_selection_animation (when selected draw an animation)

update_location (sends a signal about the towers location)

Most of the functions in tower.gd are for power management or managing when a tower was clicked or loading the assets for the towers. Generally most towers do not call any functions from tower.gd unless they are specialized functions or are requesting power.

Main unique function. Every tower has a power bank. For example the miner tower has a power bank of 1 because it does not need to store much energy but it does need to store at least one. This makes it really easy to implement the battery tower. Now, power capacity isn't held by each group, it is contributed to the group as whole. That power capacity of one doesn't belong to the miner, it is added to the group power capacity. This is the groups "bank". Any tower that generates power only makes that banks number larger. Since each bank has a maximum capacity this makes it so that we never generate infinite energy

Upgrades

Tower upgrades function on a tree structure, all upgrade info for each tower is stored in the upgrade.gd file in the folder said tower is in. When upgraded the upgrade tree is updated so the 'base' is the next node in the tree.

Solar

Main function is to generate energy, has 3 ties that organize how much energy the tower generates per 'tick' Always making sure that we don't generate more energy than we have capacity for

Battery

All this does is increase the maximum power capacity for the group. It has no other function as increasing the capacity makes it so the whole group has access to a battery of a sort.

Laser

3 types, Normal, Thel, and Pulsar. Pulsar shoots multiple very quick shots. Thel shoots less shots as it requires more power but the shots are significantly more powerful.

Missile Launcher

Same functionality as the laser but instead of shooting lines and using power, consumes minerals on each shot. Shoots 'missiles' that hone in on a target. (The never miss as they will chase an enemy until it hits it)

Power Node

No functionality. Only servers to connect towers for power. The way we set up power makes it so this node does not need to 'do' anything.

Miner

Main functionality. Look in collision radius and identify any asteroids that have minerals left. Multiple miners can target the same asteroid. Slowly drain asteroid of minerals by calling a mine function on the target asteroid.

Repair Station

Repairs any node in vicinity of it. Does this overtime in ‘bursts’